

06_implicit_anchor

```
source("R/00_setup.R")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
model_coords <- readRDS("model_coords.rds")
coords       <- readRDS("country_wave_coords.rds")
clsf         <- readRDS("classification.rds") %>% select(country = iso, class)

cat("condition labels present:", paste(unique(model_coords$condition), collapse = ", "), "\n")
```

condition labels present: C0, C1, C2

```
#' Waves are unevenly spaced, so a lag in years could reflect nothing more than
#' long gaps between fieldwork rounds. The year spine assigns each country-wave
#' its modal fieldwork year; the wave-step version at the end is the robustness
#' check against exactly that confound.
```

```
spine <- read.csv("c2_year_spine.csv") %>%
  transmute(country = as.character(country), wave = as.numeric(wave), year = as.numeric(year))
```

```

gt <- coords %>%
  transmute(country = iso, wave = as.numeric(wave), gt1 = Dim1, gt2 = Dim2) %>%
  left_join(spine, by = c("country", "wave"))
stopifnot(!any(is.na(gt$year)))

#' Distance from each model's unprompted placement to every one of that
#' country's true wave positions.

c1 <- model_coords %>% filter(condition == "C1") %>%
  group_by(model, country) %>%
  summarise(m1 = mean(Dim1), m2 = mean(Dim2), .groups = "drop")

anchor <- c1 %>% inner_join(gt, by = "country") %>%
  mutate(dist = sqrt((m1 - gt1)^2 + (m2 - gt2)^2))

```

Warning in inner_join(., gt, by = "country"): Detected an unexpected many-to-many relationship between `x` and `y`.

i Row 1 of `x` matches multiple rows in `y`.

i Row 1 of `y` matches multiple rows in `x`.

i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

```

#' ## Nearest wave and lag in years

```

```

nearest <- anchor %>%
  group_by(model, country) %>%
  summarise(anchor_wave = wave[which.min(dist)],
            anchor_year = year[which.min(dist)],
            recent_wave = max(wave),
            recent_year = year[which.max(wave)],
            n_waves = n(), .groups = "drop") %>%
  mutate(lag_years = recent_year - anchor_year,
         anchored_recent = anchor_wave == recent_wave) %>%
  left_join(clsf, by = "country")

moved <- nearest %>% filter(class != "stable")

describe_anchor <- function(df, label) {
  cat(sprintf("\n=== implicit anchor and lag, %s ===\n", label))
  df %>% group_by(model) %>%
    summarise(n = n(), median_lag = median(lag_years), mean_lag = round(mean(lag_years), 2),

```

```

        frac_anchored_recent = round(mean(anchored_recent), 2), .groups = "drop") %>%
    print()
}

describe_anchor(nearest, "all 40 countries")

```

=== implicit anchor and lag, all 40 countries ===

```

# A tibble: 4 x 5
  model          n median_lag mean_lag frac_anchored_recent
<chr>      <int>      <dbl>   <dbl>          <dbl>
1 Qwen/Qwen3.6-Plus    40         0     4.55            0.6
2 claude-opus-4-8     40         6     6.65            0.4
3 gemini-3.6-flash    40         0     4.4             0.58
4 gpt-5.5             40        4.5     5.72            0.45

```

```

describe_anchor(moved, "moved countries only (anchor is meaningful)")

```

=== implicit anchor and lag, moved countries only (anchor is meaningful) ===

```

# A tibble: 4 x 5
  model          n median_lag mean_lag frac_anchored_recent
<chr>      <int>      <dbl>   <dbl>          <dbl>
1 Qwen/Qwen3.6-Plus    36         0     4.69            0.58
2 claude-opus-4-8     36         6     7.03            0.36
3 gemini-3.6-flash    36         0     4.53            0.56
4 gpt-5.5             36        5.5     6              0.42

```

```

cat("\n=== which wave gets anchored to, country counts, moved only ===\n")

```

=== which wave gets anchored to, country counts, moved only ===

```

moved %>% count(model, anchor_wave) %>%
  pivot_wider(names_from = anchor_wave, values_from = n, values_fill = 0) %>% print()

```

```

# A tibble: 4 x 5
  model          `4`    `5`    `6`    `7`
<chr>      <int> <int> <int> <int>

```

1 Qwen/Qwen3.6-Plus	5	2	8	21
2 claude-opus-4-8	4	8	11	13
3 gemini-3.6-flash	3	4	9	20
4 gpt-5.5	4	6	11	15

```
report_ci(boot_paired(moved, "lag_years"),
  "mean lag in years, 95% CI (clears 0 = systematic lag)", digits = 2)
```

```
=== mean lag in years, 95% CI (clears 0 = systematic lag) ===
```

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	2.64	4.64	6.97	ABOVE
claude-opus-4-8	4.89	7.03	9.28	ABOVE
gemini-3.6-flash	2.61	4.50	6.58	ABOVE
gpt-5.5	4.00	5.97	8.11	ABOVE

```
#' ## The continuous anchor curve
```

```
#'
```

```
#' Mean C1-distance to each wave *position* (0 = each country's most recent),
#' which is robust to argmin jitter when two waves sit almost equally close.
```

```
cat("\n=== mean C1-distance by wave position (0 = most recent) ===\n")
```

```
=== mean C1-distance by wave position (0 = most recent) ===
```

```
anchor %>% left_join(clsf, by = "country") %>% filter(class != "stable") %>%
  group_by(model, country) %>% mutate(rel = dense_rank(desc(wave)) - 1) %>%
  group_by(model, rel) %>%
  summarise(mean_dist = round(mean(dist), 3), n = n(), .groups = "drop") %>%
  pivot_wider(names_from = rel, values_from = mean_dist, names_prefix = "pos_") %>% print()
```

```
# A tibble: 8 x 6
```

	model	n	pos_0	pos_1	pos_2	pos_3
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	Qwen/Qwen3.6-Plus	36	0.23	0.27	0.316	NA
2	Qwen/Qwen3.6-Plus	14	NA	NA	NA	0.335
3	claude-opus-4-8	36	0.18	0.184	0.227	NA
4	claude-opus-4-8	14	NA	NA	NA	0.295
5	gemini-3.6-flash	36	0.155	0.186	0.226	NA

6	gemini-3.6-flash	14	NA	NA	NA	0.256
7	gpt-5.5	36	0.194	0.222	0.257	NA
8	gpt-5.5	14	NA	NA	NA	0.266

```
#' ## Robustness: lag in wave-steps
#'
```

#' Counting in waves rather than calendar years removes any dependence on
 #' fieldwork spacing. If the lag still clears zero here, the year-based result is
 #' not a spacing artefact.

```
step <- anchor %>%
  left_join(clsf, by = "country") %>% filter(class != "stable") %>%
  group_by(model, country) %>%
  mutate(rel = dense_rank(desc(wave)) - 1) %>%
  slice_min(dist, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  select(model, country, anchor_rel = rel)

cat("\n=== implicit anchor lag in wave-steps (0 = present), moved countries ===\n")
```

```
=== implicit anchor lag in wave-steps (0 = present), moved countries ===
```

```
step %>% group_by(model) %>%
  summarise(median_steps = median(anchor_rel), mean_steps = round(mean(anchor_rel), 2),
            frac_present = round(mean(anchor_rel == 0), 2), .groups = "drop") %>% print()
```

```
# A tibble: 4 x 4
  model          median_steps mean_steps frac_present
  <chr>          <dbl>         <dbl>         <dbl>
1 Qwen/Qwen3.6-Plus      0          0.69          0.58
2 claude-opus-4-8        1          0.97          0.36
3 gemini-3.6-flash       0          0.67          0.56
4 gpt-5.5                1          0.89          0.42
```

```
report_ci(boot_paired(step, "anchor_rel"),
  "mean wave-step lag, 95% CI (clears 0 = anchors before present)", digits = 2)
```

```
=== mean wave-step lag, 95% CI (clears 0 = anchors before present) ===
```

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	0.39	0.69	1.03	ABOVE
claude-opus-4-8	0.69	0.97	1.25	ABOVE
gemini-3.6-flash	0.39	0.67	0.94	ABOVE
gpt-5.5	0.61	0.89	1.19	ABOVE

```
saveRDS(list(anchor = anchor, nearest = nearest, step = step), "implicit_anchor.rds")
```